

Opportunistic Screening: *Radiology* Scientific Expert Panel

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Radiologic tests often contain rich imaging data not relevant to the clinical indication. Opportunistic screening refers to the practice of systematically leveraging these incidental imaging findings. Although opportunistic screening can apply to imaging modalities such as conventional radiography, US, and MRI, most attention to date has focused on body CT by using artificial intelligence (AI)-assisted methods. Body CT represents an ideal high-volume modality whereby a quantitative assessment of tissue composition (eg, bone, muscle, fat, and vascular calcium) can provide valuable risk stratification and help detect unsuspected presymptomatic disease. The emergence of “explainable” AI algorithms that fully automate these measurements could eventually lead to their routine clinical use. Potential barriers to widespread implementation of opportunistic CT screening include the need for buy-in from radiologists, referring providers, and patients. Standardization of acquiring and reporting measures is needed, in addition to expanded normative data according to age, sex, and race and ethnicity. Regulatory and reimbursement hurdles are not insurmountable but pose substantial challenges to commercialization and clinical use. Through demonstration of improved population health outcomes and cost-effectiveness, these opportunistic CT-based measures should be attractive to both payers and health care systems as value-based reimbursement models mature. If highly successful, opportunistic screening could eventually justify a practice of standalone “intended” CT screening.

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According to the World Health Organization, the purpose of screening is to identify people in an apparently healthy population who are at higher risk of a health problem or condition so that early treatment or intervention can be offered and thereby reduce the incidence and/or mortality of the health problem within the population. In general medical practice, the term *opportunistic screening* refers to exercising prevention through an unorganized program or chance encounter. Within the realm of radiology, opportunistic screening has more recently been used to describe the practice of systematically leveraging imaging data that are incidental to the clinical indication for obtaining the study. A prime example of opportunistic screening that will serve as the case study for this overview is quantitative assessment of body composition data generated from CT imaging of the abdomen or thorax. By systematically quantifying aspects of muscle, fat, bone, liver, arterial calcification, and other structures, valuable prognostic data may be used for risk stratification or presymptomatic detection of relevant disease (1,2). Although opportunistic screening can apply to other imaging modalities, such as conventional radiography, US, and MRI, most attention to date has focused on body CT and, more recently, by using artificial intelligence (AI)-assisted methods (3–7). This growing interest in population-based opportunistic CT screening can be attributed to a confluence of multiple superimposed factors, including (a) the objective nature of CT-based measurements, (b) the standard inclusion of the entire torso cross-section for body imaging, (c) the relative data reproducibility across CT vendors, (d) the large number of CT scans performed each year (which also speaks to its clinical value), (e) the emergence of “explainable” AI

algorithms to replace the need for more onerous manual or semiautomated tools that were rarely performed in routine practice, and (f) the overall increased interest in adding value to routine radiology practice.

That CT images (as well as those of other imaging modalities) are inherently data rich beyond their often-narrow clinical indication (eg, evaluation for diverticulitis) can have both favorable and unfavorable consequences. Until recently, the perceived negative aspects of finding “incidentalomas” had garnered the most attention (8,9). Often overshadowed by this concern for potential harms related to incidental imaging findings are the potential benefits. When leveraged to full advantage, one could argue that the cumulative effect related to incidental CT imaging data may confer a net benefit to patients (1). Cardiovascular disease, osteoporosis, metabolic syndrome, sarcopenia, and cancer all represent major public health issues in our diverse and aging population and account for the majority of all deaths. Detection of any of these conditions before the onset of symptoms in patients with unsuspected higher risk could translate into improved clinical outcomes, if appropriate lifestyle modifications or relevant medical interventions are implemented in a timely fashion.

As described, body CT of the chest or abdomen can provide a fairly comprehensive evaluation of tissues and organs that are often unrelated to the intended clinical indication for imaging. Specific examples of such quantitative assessment (Figs 1, 2) include bone mineral density (for osteoporosis), visceral fat (for metabolic syndrome), muscle (for sarcopenia), liver (for steatosis), and arterial calcification (for atherosclerosis). Beyond these measures of body composition, other potential organ-based assessments are

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Abbreviations

AI = artificial intelligence, CAdE = computer-aided detection, CPT = Current Procedural Terminology, DXA = dual x-ray absorptiometry, FDA = Food and Drug Administration, MIMPS = medical image management and processing systems.

Summary

Within radiology, opportunistic screening entails leveraging potentially valuable imaging data incidental to the clinical indication for imaging.

Essentials

- Systematic use of quantitative imaging findings generally unrelated to the clinical indication can be used for population health purposes, including risk profiling and presymptomatic detection of relevant disease.
- The objective, reproducible, and efficient automation provided by artificial intelligence (AI) algorithms likely represents a critical step for the widespread implementation of opportunistic CT screening.
- The clinical success of systematic opportunistic screening hinges on acceptance by radiologists, referring providers, and patients.
- Regulatory clearance, demonstration of cost-effectiveness, and reimbursement are important considerations that must all be adequately addressed for the successful commercialization and clinical implementation of AI-assisted opportunistic screening.

feasible, such as opportunistic detection of hepatic fibrosis and cirrhosis, diabetes mellitus, and urolithiasis (Fig 3). Experience with intended CT screening colonography has demonstrated that the detection of unsuspected osteoporosis, abdominal aortic aneurysms, and extracolonic cancers can positively impact both the clinical efficacy and cost-effectiveness of the examination (10–12). Of course, these incidental observations can also broadly apply to all abdominal (and chest) CT. Emerging evidence suggests that opportunistic CT screening for osteoporosis and cardiometabolic diseases using AI-assisted body composition tools may override the usual practice of ignoring such data, in that it improves clinical efficacy and is more cost-effective (actually with cost savings) compared with ignoring these findings (13). Furthermore, an innovative practice of intended CT screening (ie, not opportunistic) using these same quantitative measures might ultimately prove to be cost-effective as well.

Early investigations in opportunistic CT-based body composition assessment were limited to manual and semiautomated methods for assessing bone mineral density, aortic calcium, abdominal fat, skeletal muscle, and liver fat. Nonetheless, these nonautomated assessments provided proof of concept of their

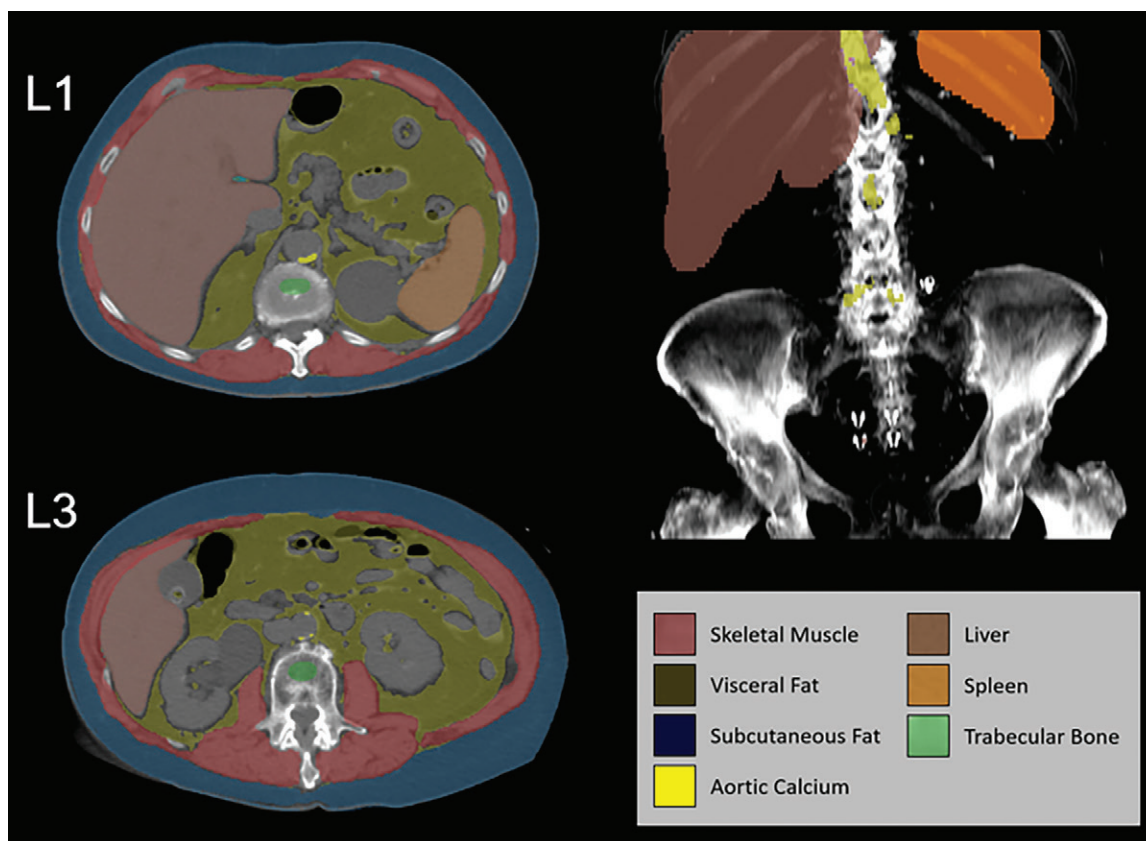


Figure 1: Case example of artificial intelligence (AI)-assisted cardiometabolic opportunistic screening in a 77-year-old woman with flank pain who underwent abdominal CT for urolithiasis evaluation. Automated CT-based body composition algorithms can be applied (prospectively or retrospectively) regardless of the clinical indication for imaging. Unenhanced transverse CT scans at the L1 (top, left) and L3 (bottom, left) vertebral levels and a coronal maximum intensity projection image for volumetric parameters (top, right) show the automatic segmentation of skeletal muscle, abdominal fat (visceral and subcutaneous), trabecular bone, aortic calcium, liver, and spleen (see color legend) provided by this particular research-based suite of AI tools. In this case, unsuspected osteoporosis (L1 trabecular attenuation <100 HU), sarcopenia (L3 muscle attenuation of 20 HU), and advanced atherosclerotic disease (aortic Agatston score of 5889) are evident when these tools are retrospectively applied (note that this scan was obtained more than a decade ago). Without prospective reporting, this patient went on to have an osteoporotic vertebral compression fracture, in addition to subsequent acute myocardial infarction that led to heart failure and death.

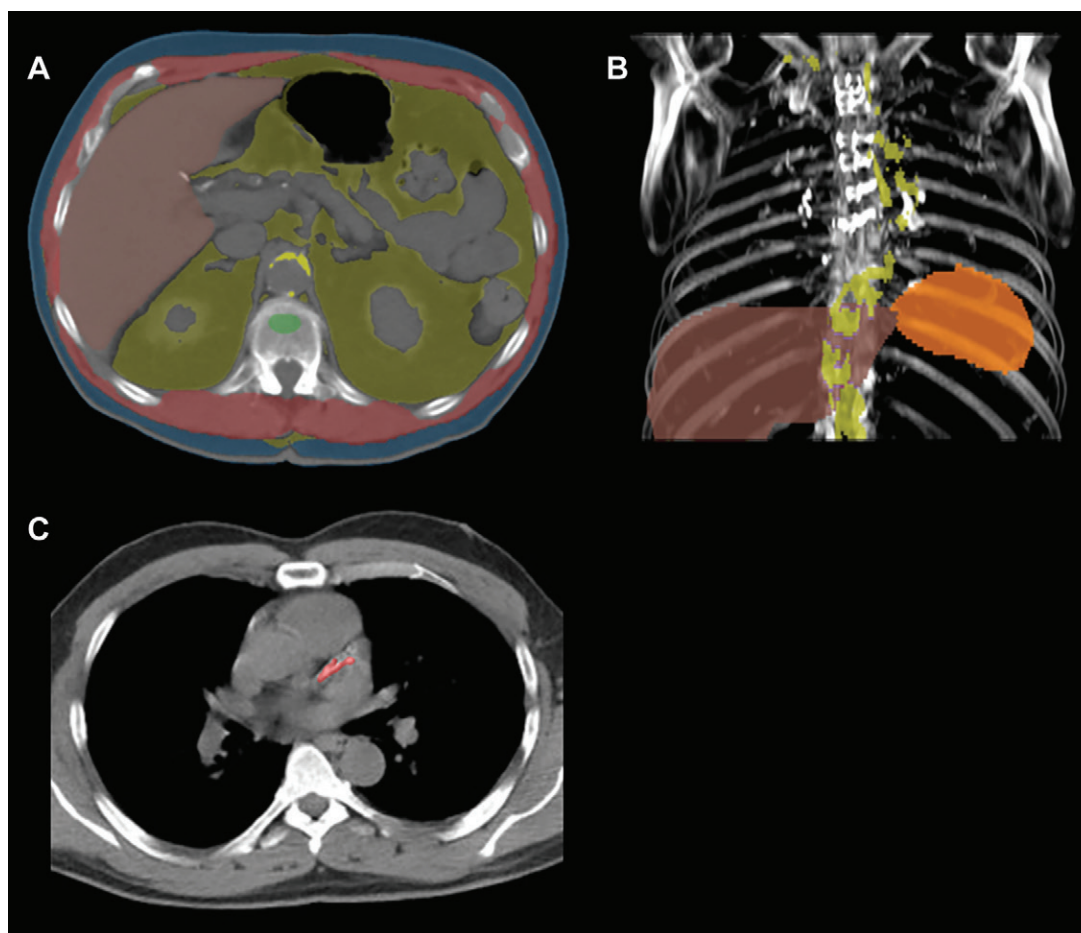


Figure 2: Artificial intelligence (AI)-assisted cardiometabolic opportunistic screening at chest CT. **(A)** Unenhanced transverse CT scan at the T12 vertebral level and **(B)** coronal maximum intensity projection image with volumetric data show automatic segmentation and quantification of body composition data, similar to the abdominal example in Figure 1. **(C)** Unenhanced CT scan shows nongated coronary calcium segmentation and scoring. Other opportunistic cardiopulmonary measures are possible, including aortic calcific plaque.

prognostic value. Examples include manual vertebral trabecular attenuation values for assessing bone mineral density and semiautomated quantification of aortic calcification for cardiovascular risk (14–16). Other examples include liver attenuation measurement and intra-abdominal fat quantification for hepatic steatosis and metabolic syndrome (17,18). The development of fully automated “explainable” AI body composition tools to supplant these analogous but tedious manual or semiautomated methods represents a major emerging breakthrough for opportunistic screening (Figs 1–3) (19–26). Emerging data acquired using fully automated body composition tools have demonstrated the population-based potential for predicting downstream adverse clinical outcomes, including major cardiovascular events, osteoporotic fractures, and death (3,5,6).

Challenges to Clinical Implementation

Some remaining requisites to broader clinical adoption and implementation of CT-based opportunistic screening include widespread availability of emerging fully automated AI algorithms to replace labor-intensive manual measurements; generalization of results to more racially, ethnically, and socioeconomically diverse patient cohorts; and demonstration of reproducibility

across disparate technical settings. Research groups, such as the multi-institutional collaborative research consortium known as OSCAR (Opportunistic Screening Consortium in Abdominal Radiology), are gathering more numerous and diverse data that should allow for more robust risk stratification models. Use of retrospective CT data allows for built-in long-term clinical follow-up for subsequent adverse events that would otherwise require a much more expensive and prolonged prospective study design. Other challenges to the implementation of opportunistic screening are discussed hereafter.

AI Acceptance by Radiologists, Referring Providers, and Patients

The AI revolution is upon us. From personal assistants to self-driving vehicles, the public’s acceptance of AI is unmistakable as AI-driven tools are seamlessly incorporated into nearly every aspect of daily life. However, in the medical domain, the public’s understanding of the emergence and use of AI tools is less clear. Radiology, in particular, has benefited from advances in image recognition and processing, two areas where great strides have been made in the application of machine learning algorithms. The explosive growth and reliance on CT imaging, most notably

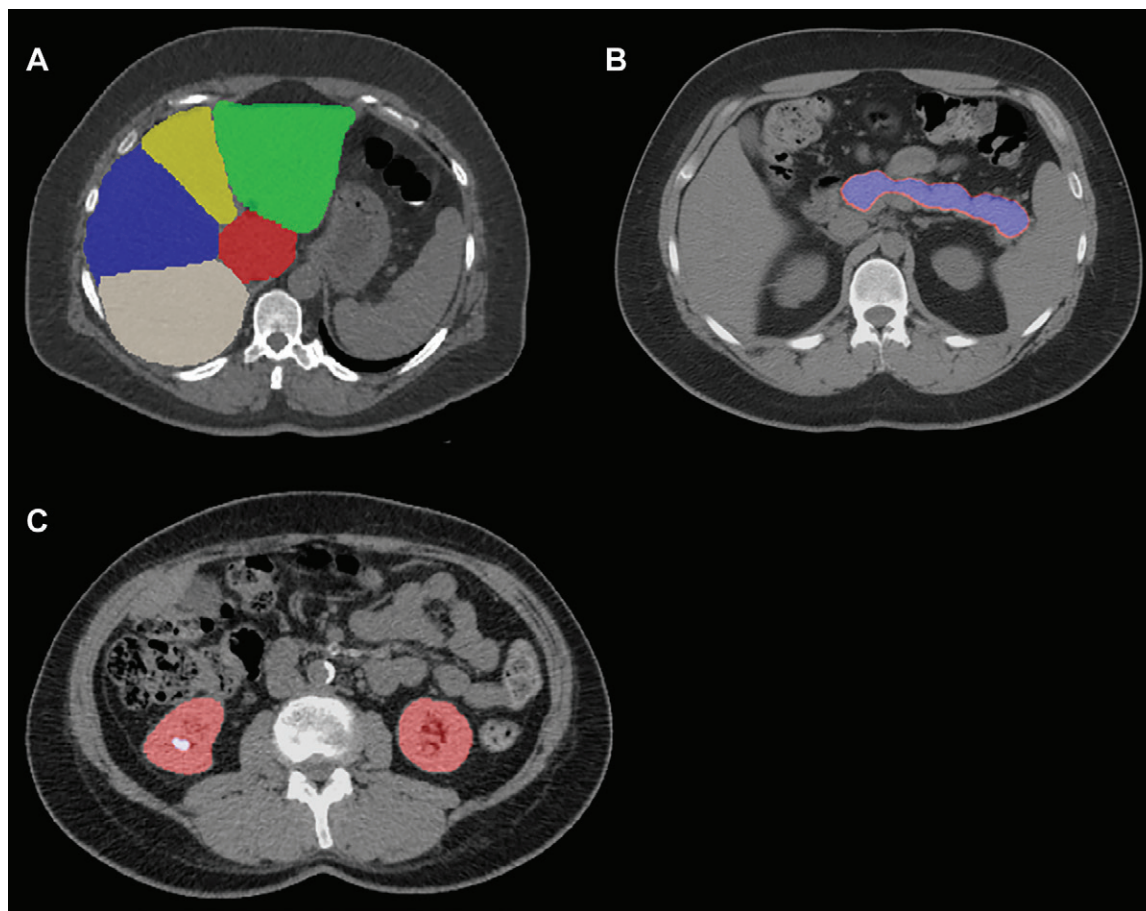


Figure 3: Examples of other automated opportunistic assessments in three different patients. **(A)** Axial unenhanced CT scan shows automated segmentation of the liver into Couinaud segments, which allows for assessment of segmental redistribution and can demonstrate unsuspected hepatic fibrosis. **(B)** Axial unenhanced CT scan shows automated segmentation of the pancreas, which can help screen for changes suggestive of underlying diabetes. **(C)** Axial unenhanced CT scan shows automated segmentation of the kidneys and renal calculi (right kidney), which allows for stone detection, as well as quantitative volumetric assessment of renal size and stone burden.

over the past 2 decades, has resulted in a wealth of digital data, creating an opportunity to apply machine learning algorithms to derive even greater knowledge from the data acquired from imaging, opening the door to opportunistic screening. The question is whether imaging stakeholders (radiologists, referring providers, and patients) are willing to embrace this additional knowledge or be overwhelmed by it.

Radiologist Workload Concerns

Radiologists currently encounter long lists of unread studies, with growing pressures to interpret studies within short turnaround times alongside requests for 24-hour availability. The current radiology workforce is struggling to meet these demands, which is leading to burnout and diminished well-being (27). The prospect of adding more information for radiologists to convey in their reports, particularly opportunistic findings not requested by the referring provider, has the potential to be rejected by the radiology community. However, if this additional information can be obtained and incorporated into reports in a fully automated fashion, the likelihood of success should be much higher. Radiologists will also need to be reassured of an adequate reimbursement to offset the expense of

incorporating these opportunistic screening AI technologies. If both of these aforementioned obstacles are overcome, the potential for routine reporting of opportunistic information by radiologists is more likely to be embraced and owned by the radiology community. By pursuing and routinely incorporating information gleaned from opportunistic screening, radiologists can position themselves as drivers of population health. Furthermore, although it is not a requirement that radiologists control the development and use of imaging-based opportunistic data, they are clearly in the best position to understand the quality assurance process to promote best practices. If opportunistic imaging data can further be incorporated with additional data streams housed in pathology, genetic, and laboratory data warehouses, then radiologists can uniquely position themselves at the center of modern health care.

More Work for Referring Providers

Referring providers may also represent obstacles to acceptance, even though the additional information gleaned from imaging could be beneficial to their patients. In particular, primary care physicians who order a large number of CT examinations could become responsible for this additional opportunistic

information that they did not request. A finding of osteoporosis, hepatic steatosis, or increased visceral fat, while valuable information, may require follow-up by primary care providers, such as additional consults or modification of medical management. If such steps are not pursued, any potential benefit to the patient is lost. In addition, these referring providers may fear medical liability related to any perceived lack of follow-up. Providers will also likely face additional questions from their patients regarding these incidental findings, adding to their already large burden of being available to their patients. To reduce the need for excessive oversight and explanation, referring physicians will rely on the radiology community to create reporting styles that allow patients to better understand the implications of unsuspected cardiometabolic findings.

Higher Costs, Medical Jargon, and Poor Referral Networks for Patients

From a patient perspective, opportunistic screening could improve the value of an imaging encounter, if incorporated thoughtfully. Value in health care can be defined as outcome improvement relative to the expense to achieve those outcomes. Unfortunately, in the United States, patients rarely receive high-value care, and costs borne by patients continue to increase due to increasing deductibles and greater cost-sharing. Imaging, in particular, represents a large percentage of out-of-pocket expenses (28). Opportunistic screening that delivers on the promise of predictive data for better risk stratification or early detection of unsuspected disease beyond the indication for imaging may help to rebalance the value equation in favor of patients. However, this information needs to be delivered in a readily digestible format. The current radiologic report is designed to serve the needs of referring providers, not patients, and is often replete with technical jargon. For patients to derive the additional value afforded by opportunistic screening, reports will need to evolve to better cater to patients (29). Moreover, more seamless referral networks must exist for patients to receive the necessary care recommended by valuable opportunistic information. For example, if a patient cannot access the necessary treatments for osteoporosis incidentally detected at CT and subsequently suffers a hip fracture, the additional information ceases to become valuable to the patient. Patients may seek direct input from radiologists as the medical professional to convey this additional information, placing the specialty in an unfamiliar role. If radiologists can leverage this encounter and create the bandwidth to accomplish this task, this may provide a future avenue for value-added contributions. Lastly, patients must be willing to undergo the necessary lifestyle modifications or medical interventions to receive any real clinical benefit of unsuspected opportunistic findings.

Technical Challenges

Extensive algorithm development is underway for opportunistic screening applications. Some areas are better established with more well-developed software than others. For example, fat segmentation has been a staple of radiologic image processing for decades. It is made easier by the excellent separation of fat and soft tissue on CT images. Fat segmentation on MRI scans is also well studied and relatively easy to perform. Muscle segmentation

is well established and performs robustly in assessing both muscle bulk and density (30). Because the majority of literature on fat and muscle quantification preceded the machine learning revolution, most analyses are performed on a single abdominal image section. This approach is easier to standardize and generally allows for reproducible and robust assessment. In the future, however, localized or whole-body analyses may become relevant if improved prediction can be demonstrated. The liver is another established area of machine learning–based quantitation, especially for opportunistic detection of diffuse processes such as steatosis and fibrosis (21). Due to the availability of high-quality well-annotated public data sets, liver CT segmentation can now be performed robustly with Dice coefficients of 0.95 for contrast-enhanced CT and 0.89 for noncontrast CT (31). Disease states like extensive liver metastases and perihepatic ascites provide segmentation challenges for which further research is needed. Manual and semiautomated region-of-interest placement for CT-based bone mineral density assessment is fairly well established (32). Fully automated bone mineral density measurement markedly increases the challenge due to scoliosis, degenerative changes, compression fractures, anatomic variants, and issues related to patient positioning (33). Atherosclerotic plaque assessment, whether in the chest or abdomen, has also been investigated for opportunistic use at routine CT, including nongated coronary artery and aortic calcium scoring. The presence of intravenous contrast, which affects quantitation of plaque, has been addressed in some studies and has the potential to further increase usage of these AI tools (34).

Opportunistic screening software will need to adapt to differences among vendors and imaging centers. CT-based AI-generated results must prove to be generalizable across various scan acquisition parameters and reconstruction kernels, section collimation and thickness, and other technical variations. For example, CT-based atherosclerotic plaque scoring is traditionally done on 3-mm sections. Fully automated calcium scoring on thicker or thinner sections will need to be accompanied by appropriate caveats or corrections. Radiation dose and kilovolt settings are other factors likely to affect the quality of AI-generated measurements. For example, CT attenuation measurements (especially for bone mineral density), plaque scoring, and organ volumetry may be affected by varying milliamperage and kilovolt levels. With both CT and MRI, artifacts may adversely affect quantitation and must be recognized by the software.

Normative data for body composition metrics are needed. Such data enable the conversion of measurements to statistical scores. This is familiar to radiologists and nuclear medicine physicians who perform dual x-ray absorptiometry (DXA). T scores and z scores are normalized assessments of bone mineral density adjusted for age and sex. By applying similar normative techniques, measurements such as fat and muscle volume and attenuation can be converted to age- and sex-specific scores that indicate deviation from normality, either on the low or high sides, expressed as a percentile. Normalization according to racial and ethnic groups may also prove to be important for some metrics. Such normalization may prove challenging for small groups but may be addressable by pooling data across institutions or countries, as with the aforementioned OSCAR (Opportunistic Screening Consortium in Abdominal Radiology) trial.

Commercial AI systems frequently prepare well-formatted graphic reports with quality control images and tabular summaries of the results. As with similar systems commonplace for echocardiography and pulmonary function testing, such results and reports need to be reviewed by the radiologist. However, with time and increasing performance of the software, AI-generated reports will likely not require further editing for the majority of studies, similar to DXA reports. Such AI-generated reports could greatly lengthen existing radiologic reports, leading to information overload for the referring physician. Structured reporting and good graphic displays may ameliorate this issue.

Regulatory Hurdles

AI applications in radiology that use images and/or pixels in their input are currently regulated by the U.S. Food and Drug Administration (FDA) as “Software as a Medical Device” (SaMD). To date, the FDA has approved approximately 250 AI and Machine Learning (AI/ML) imaging products as SaMD (35). Per the current clearance paradigm, these algorithms are “locked” prior to marketing and any updates or continuous learning would require resubmission and reapproval. Historically, the FDA has referred to AI/ML software that analyzes medical images as computer-aided detection (CADe), computer-aided diagnosis (CADx), and computer-aided triage (CADt). More recently, the FDA has further characterized this software into the following categories: computer-aided diagnosis (CADx), which aids in characterizing or assessing disease, disease type, severity, stage, or progression; computer-aided detection (CADe), which aids in localizing or marking regions that may reveal specific abnormalities; computer-aided detection and diagnosis (CADe/x), which aids in both localizing and characterizing conditions; computer-aided triage (CADt), which aids in prioritizing or triaging time-sensitive detection and diagnosis in patients; computer-aided acquisition and/or optimization (CADA/o), which aids in acquisition or optimization of images or diagnostic signals; and medical image management and processing systems (MIMPS), which are quantitative of anatomic features or functions, such as with quantification, image reconstruction, applying filters, segmentation, artifact reduction, and de-noising.

Nearly all imaging AI software as a medical device are FDA class II devices, which may achieve clearance under the De Novo pathway, or the 510(k) pathway if a predicate device exists. When the FDA evaluates a product for clearance, a benefit-risk analysis is performed to ascertain that the probable health benefits from the use of the device outweigh any probable injury or illness from such use. This analysis would then determine the type of “special controls” required (eg, standalone performance testing, clinical reader studies). Body composition algorithms would likely fall under either a CADe or MIMPS clearance, depending upon the specific outputs produced. For example, in evaluating the liver, a tool could return the mean attenuation value (in Hounsfield units), which would be a quantification tool (MIMPS) replacing the manual radiologist task of placing a region of interest on the liver. Alternatively, the tool could evaluate the liver for mild, moderate, or severe hepatic steatosis, which would be the detection of disease pathology (CADe). The distinction between these two may be hazy. For example, current

commercial tools for evaluating coronary calcium still return semiquantitative categorizations such as low, medium, and high, which could be construed to mean mild, moderate, and severe. Looking at the current market landscape, all the use cases similar to opportunistic AI (eg, coronary artery calcium scoring or fatty liver scoring at MRI) are currently approved as postprocessing software devices.

The level of evidence required for a CADe clearance is generally a much higher bar compared with that of a MIMPS clearance. For devices using computer-aided detection or diagnosis, the FDA will typically require clinical validation studies, including standalone performance testing (eg, evaluating sensitivity and specificity by running the algorithm on a statistically significant sample of cases), as well as clinical reader studies (comparing the performance of radiologists aided and unaided by the software). For MIMPS tools, typical bench testing for reproducibility or phantom studies may be sufficient.

One of the major challenges for commercialization and widespread adoption of AI-assisted opportunistic screening revolves around integration into the current workflow of radiology departments. Because these are not tasks currently undertaken by radiologists, it is unlikely that already overburdened physicians will be eager to assume additional time-consuming tasks. Minimizing technical deployment hurdles (eg, by deploying a platform with existing infrastructure) would likely help with adoption. Additionally, seamless integration with the current reporting workflow would also likely be helpful. While some AI tools on the market today produce a separate portable document format (PDF) report, it is unclear where such a report should be housed within the electronic health record and how the radiologist would interact with it. Instead, if the values from the algorithms were automatically populated within the radiologic report, then these outputs would become an integral part of the radiologist’s work product, with minimal effort on the part of the radiologist.

Payment Models for Reimbursement

Reimbursement opportunities for opportunistic screening vary by country depending on whether the country pays for services at the population level (eg, most single-payer systems) or at the service level (eg, fee for service). The U.S. health system is diverse, with specific regions behaving like single-payer systems (eg, integrated health systems such as Kaiser Permanente), while others are predominantly fee-for-service models furnished by multiple payers. The general mechanism of payments for clinical services in the outpatient setting in the United States is codified with Current Procedural Terminology (CPT). CPT codes are created and maintained by the CPT editorial panel convened by the American Medical Association. Three categories of codes exist under the CPT umbrella. Category 1 CPT codes are frequently paid by Medicare and commercial payers. Category 3 CPT codes are typically created for tracking emerging technology and are not routinely paid across all payers. Category 2 CPT codes are used for reporting performance measures, reducing the necessity for chart review and medical records abstraction. Category 2 codes are not associated with fee-for-service payment but may have an impact

on payments made in value-based contracts that reimburse clinicians who score favorably on reporting and performance.

Several common requirements are shared between category 1 and 3 CPT codes. For example, the service descriptor is unique, well defined, and describes a procedure or service that is clearly identified and distinguished from existing procedures and services already in the CPT. Second, the proposed descriptor for the procedure or service is not a fragmentation of an existing procedure or service, or currently reportable as a complete service by one or more existing codes. For example, the CPT editorial panel would not allow a CPT code for measuring lymph nodes on a chest CT scan as that is assumed to be part of the payment already captured for the currently existing CPT code for chest CT. However, the panel may view an opportunistic screening service at that same chest CT examination differently if the service is something not typically provided, especially if the service needs advanced postprocessing or AI inference beyond the scope of human capabilities. Select radiologic opportunistic screening applications are likely to meet these two and other required criteria for category 1 and 3 CPT codes. For the best chance of payment, opportunistic screening in radiology would ideally meet all four of the following unique category 1 status criteria (36): (a) All devices (including software) and drugs necessary for the performance of the procedure or service have received FDA clearance or approval when such is required for the performance of the procedure or service. (b) The procedure or service is performed by many physicians or other qualified health care professionals across the United States. (c) The procedure or service is performed with a frequency consistent with the intended clinical use (ie, a service for a common condition should have high volume). (d) The procedure or service is consistent with current medical practice.

FDA clearance and literature documentation will likely be the highest barrier to achieving such a status. One example of a recently created category 1 CPT code fitting into the general domain of opportunistic screening is the trabecular bone score (37). The trabecular bone score is generated by software analyzing previously acquired DXA scans, quantifying the trabecular bone in a specific field of view to allow for analysis of osteoporosis risk. The CPT code set is considered an add-on, meaning they are billed in addition to the primary procedure (DXA). Medicare and some commercial payers are honoring the category 1 status for trabecular bone score and routinely paying for its use. However, achieving category 1 CPT status does not always translate into payment. Examples of screening imaging examinations meeting category 1 criteria but not covered by Medicare national coverage determinations include CT colonography and coronary calcium scoring, demonstrating the vagaries of the U.S. health care system. The exact reasons for having a denial of payment even if a service has achieved a category 1 CPT code are varied, nuanced, and sometimes buried in politics that are beyond the scope of this panel review.

If category 1 CPT status cannot be achieved, the lower barrier for category 3 codes requires that the procedure or service is currently or recently performed in humans and at least one of the following additional criteria has been met (36): (a) The application is supported by at least one CPT or Health Care

Professional Advisory Committee representing practitioners who would use this procedure or service; or (b) the actual or potential clinical efficacy of the specific procedure or service is supported by peer-reviewed literature for examination by the CPT editorial panel, or at least one institutional review board–approved protocol of the procedure or service exists, or a description of a current and ongoing U.S. trial outlining the efficacy of the procedure or service exists, or other evidence of evolving clinical use exists.

An example of an opportunistic screening imaging application that has received category 3 CPT status is the detection of vertebral compression fracture. This code is an add-on to a primary service (CT) offering further analysis for early or pending vertebral body compression deformities. The code is not currently associated with payment by most payers in the U.S. health system. However, achieving category 3 status does allow for tracking of the service, which can help future bids for achieving category 1 status (38). As noted, most category 3 codes are not associated with payment from Medicare or other U.S. insurers.

As mentioned, category 2 CPT codes are not associated with fee-for-service payments but are used for tracking performance in Medicare pay-for-performance programs. Although no specific measures currently exist that would be optimized by opportunistic screening at imaging examinations, a few current measures could be relevant after minor adjustments of the category 2 descriptors and intent. For example, a category 2 code currently exists for tracking the performance of osteoporosis screening with DXA. Minor modifications could be made to allow for the expansion of screening opportunities for osteoporosis using quantitative CT-based algorithms for bone mineral density assessment and detection of vertebral compression deformities. This expanded definition will allow a clinician to meet the performance criteria by including not only patients who have received DXA but also patients who had CT imaging for unrelated reasons. This could potentially expand the overall numerator of patients used in calculating performance on this measure, translating into a higher pay-for-performance payment.

Several other rare mechanisms of capturing payment for opportunistic screening applications exist inside U.S. health care payment policy. One worth mentioning is the New Technology Ambulatory Payment Classification (APC). The New Technology APC is independent of previously mentioned coding systems and is used solely to pay hospitals for new technological outpatient services (a New Technology Add on Payment [NTAP] also exists for inpatient services). The NTAP and New Technology APC have a similar but independent level of evidence criteria as that of category 1 and 3 CPT codes, but the details are beyond the scope of this article. Examples of services receiving NTAP or New Technology APC payment for imaging-based opportunistic screening applications are lacking. However, a loose comparison could be drawn between quantitative analysis of the coronary arteries with HeartFlow (<https://www.heartflow.com>) (39). HeartFlow is an interactive revascularization CT fractional flow reserve planner tool. Using HeartFlow, a coronary CT angiogram is analyzed using advanced postprocessing techniques to predict the severity of coronary artery stenosis by computing flow dynamics beyond what is capable with human interpretation. This application has shown outcome benefits and has qualified for

a New Technology APC. Opportunistic screening applications that perform advanced quantification beyond the capabilities of a radiologist and prove outcome benefits may qualify for such payments in the future.

The current review is not exhaustive of all possible payment mechanisms, but instead represents a targeted look at the most frequent and desirable avenues toward reimbursement in fee-for-service systems. All the mentioned payment mechanisms will require, at a minimum, research with evidence of impact on patient care, costs, and outcomes, as well as assurance that the application is not redundant with work a radiologist could reasonably perform without the application.

Reimbursement in systems that pay at the population level (integrated systems in the United States and single-payer systems globally) hinges on providing services to patients that improve the health of the population, while keeping costs level or decreasing. In the United States, health systems that participate in risk-sharing or are fully integrated often shift focus from treating acutely sick patients to encouraging wellness and population health initiatives, including screening programs similar to single-payer systems. Opportunistic screening can help uncover patient-level risk factors that could lead to serious disease in the future. Early intervention in managing such risk factors would hopefully lead to better patient outcomes and affordable overall cost of care. Opportunistic applications that provide such improvements in patient outcomes at a lower or neutral cost would likely be adopted without any direct payment vehicle, such as achievement of CPT code categories 1–3.

Conclusion

Systematic opportunistic screening offers great potential to add value to the imaging services that radiologists already provide to their patients. However, as discussed in this review, there are challenges and barriers to clinical implementation that must first be adequately addressed.

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